UNIVERSITY OF WESTMINSTER

SCHOOL OF COMPUTER SCIENCE
AND ENGINEERING

Module Title: **Mathematics for Computing**
Module Code: **4COSC002W.1**
Exam Period: **15 December 2023**
Time allowed: **90 minutes**

INSTRUCTIONS FOR CANDIDATES

- Students who arrive more than 30 minutes late will not be permitted to enter the examination room and must apply for Mitigating Circumstances.
- Exiting the examination room is not allowed during the first 30 minutes and the last 15 minutes of the exam period.
- You are advised (but not required) to spend the first ten minutes of the examination reading the questions and planning how you will answer them.
- Students are not allowed to visit the toilet unaccompanied. If you need a comfort break during the examination, please be escorted by a tutor. We ask everyone to minimise disturbances when moving in and out of rows.
- Calculators may be used provided they are silent, cordless, not pre-programmed by the candidate and cannot receive or transmit data remotely.
- It is mandatory to answer ALL QUESTIONS. The marks allocated for each task are indicated within the task's description.

The following information is given:

Stats Basics Formulae – Appendix 1

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IN-CLASS TEST 2

THIS PAPER MUST NOT BE TAKEN OUT OF THE EXAMINATION ROOM

DO NOT TURN OVER THIS PAGE UNTIL THE INVIGILATOR INSTRUCTS YOU TO DO SO

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PART A – SHORT ANSWERS

(Provide concise, direct answers to each question. No additional explanations or workouts should be provided)

QUESTION A1 (5 MARKS)

Fill the gaps in the given statements:

A. A matrix that has 1's on the diagonal and 0's elsewhere is called an _____ matrix.

B. A matrix where all elements off the main diagonal are zero, but the diagonal itself can contain non-zero elements, is known as a _____ matrix.

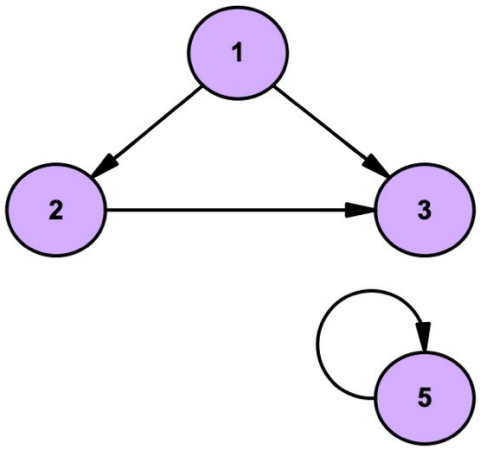
C. The probability of rolling a specific number on a six-sided die is an example of _____ probability.

D. A graph is connected if for every pair of nodes, there is a _____ between them.

E. The depth of a node in a tree is the number of _____ from the root to that node.

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QUESTION A2 (8 MARKS)



For the following **graph N** represented above, do the following:

A. form graph notation for the given graph N: a set **V** of all its nodes and set **E** of all its edges;
B. determine if it is a *directed* or *undirected* graph;
C. determine if it is a *connected* or *disconnected* graph;
D. determine if it is a *cyclic* or *acyclic* graph.

QUESTION A3 (7 MARKS)

You are given the following equation, where X is an unknown matrix:

$$3X - \begin{pmatrix} 2 & 5 \\ 3 & 6 \end{pmatrix} = 7X + \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}$$

Find the values of X_{11} and X_{22} in the matrix X that satisfy this equation.

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QUESTION A4 (8 MARKS)

Given matrices A and B:

$$A=\begin{pmatrix}6 & -3 & 2\\ 4 & 5 & -1\end{pmatrix} \text{ and } B=\begin{pmatrix}-2\\ 3\\ 1\end{pmatrix}$$

Calculate the matrix C such that $C=A\times B$.

Identify the elements C_{11} and C_{21} in your answer, where the first index indicates the row and the second index indicates the column.

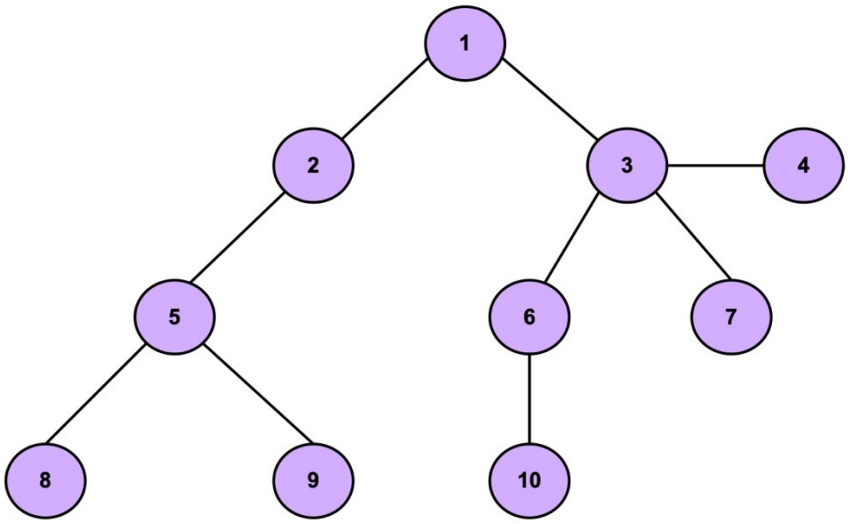
QUESTION A5 (6 MARKS)

A software company has developed 100 programs, with the following distribution:
- 25 programs written in Python.
- 30 programs written in Java.
- 20 programs written in C++.
- The remaining programs are written in JavaScript.

Find the probability of selecting a program that is NOT written in Java.

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QUESTION A6 (6 MARKS)



Tree H is given above.

Identify:

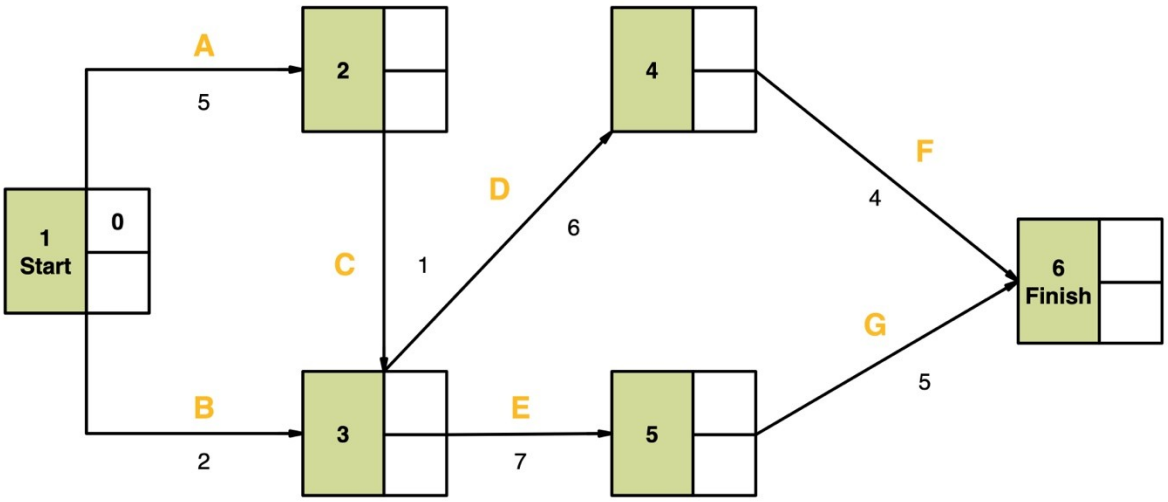
A. all the leaves of a tree;
B. depth of a tree;
C. is the given tree a binary tree (yes/no)?

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PART B – DETAILED WORKOUTS

(Provide a comprehensive, step-by-step workout of each question. Each step should include relevant formulas, calculations, and brief explanations where necessary to justify your approach).

QUESTION B1 (15 MARKS)



The provided diagram represents a project schedule using a network graph, where each node (or box) represents a task within the project. The edges represent activity and are labelled with letters, and the duration of each task is given below. The arrows indicate the sequence of tasks, demonstrating which tasks must be completed before others can begin.

Using the provided network graph, your objective is to perform a critical path analysis. This will involve identifying the longest path through the network and determining when the project can be completed.

For your analysis, follow these steps:

A. Draw the provided network graph on your answer booklet. Fill in all the empty cells (EST and LFT) for each node.
B. Provide the calculations of the Earliest Start Time (EST, the upper empty cell of the node) and the Latest Finish Time (LFT, the lower empty cell of the node) for each task.
C. Identify the critical path, which is the path from Start to Finish that has zero slack for all tasks. Show the critical path on your drawn graph. If there is more than one critical path, name at least one.

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QUESTION B2 (13 MARKS)

A fair die is rolled twice, yielding two outcomes: $X1$ (the result of the first roll) and $X2$ (the result of the second roll). Consider two events:

Event A: $X1=5$ or $X2=5$ (at least one of the rolls results in a 5).
Event B: $X1+X2=8$ (the sum of the two rolls is 8).

Find the conditional probability $P(A \vee B)$, which is the probability of Event A occurring given that Event B has occurred.

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QUESTION B3 (15 MARKS)

Analyse network latency by examining the time taken for packets to travel to a server and back. The dataset below records the round-trip times (RTT) for packets in milliseconds.

Dataset: Packet Round-Trip Times (in milliseconds)

Packet	RTT (ms)
P1	45
P2	60
P3	45
P4	75
P5	60
P6	60

Calculate the following statistical measures:

A. Mean: Average RTT.
B. Range: RTT range.
C. Median: Central RTT.
D. Mode: Most frequent RTT.
E. Standard Deviation: RTT consistency.
F. Variance: Degree of RTT spread.
G. Interquartile Range (IQR): Spread of the middle 50% of RTTs.

NB: In the real ICT2 paper you will be asked to find either *Standard Deviation and Variance* (Questions E, F) or *IQR* (Question G).

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QUESTION B4 (17 MARKS)

You are provided with two different 3x3 matrices. Your task involves a comprehensive analysis of these matrices. Each matrix presents a unique case: one is singular (with a determinant of 0), and the other is non-singular (with a non-zero determinant).
Matrices Provided:

Matrix 1:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Matrix 2:

$$\begin{bmatrix} 2 & 3 & 1 \\ 5 & 7 & 4 \\ 1 & 0 & 6 \end{bmatrix}$$

Perform the following analyses for each matrix:

A. Determine the determinant of each matrix using the diagonal method. Identify if the matrix is singular or non-singular.
B. For the non-singular matrix, calculate the matrix of minors for each element.
C. For the non-singular matrix, compute the matrix of cofactors for each element.
D. For the non-singular matrix, derive the adjoint matrix from the matrix of cofactors.
E. If the matrix is non-singular, calculate its inverse.

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APPENDIX 1 – FORMULAE SHEET

Mean (Arithmetic Average)	$\acute{x}=\frac{\sum_{i=1}^{n} x_i}{n}$
Standard Deviation (for a sample)	$s=\sqrt{\sum ¿¿¿}$
Variation (for a sample)	$s^2=\sum ¿¿$
Quartiles' Positions	$\text{Q Position}=\frac{Q}{100}\times(n+1)$
Interpolation	$\text{Int. Value}=X_{\lfloor p \rfloor}+(p-\lfloor p \rfloor)\times(X_{\lfloor p \rfloor+1}-X_{\lfloor p \rfloor})$

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